

Bypass for CVE-2024-9956 in Safari on iOS

Bypass for CVE-2024-9956 in Safari on iOS - Dennis Kniep, Dennis Kniep.

- Published: 2025-09-24
- Original: <<https://denniskniep.github.io/posts/13-bypass-cve-2024-9956/>>
- Preserved from: <https://denniskniep.github.io/posts/13-bypass-cve-2024-9956/> (live) on 2026-09-29
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Content

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TL;DR;

I found a bypass for the recently fixed vulnerability, CVE-2024-9956, in Mobile Safari. The original fix blocks `FIDO:/` URIs from being navigable. I was able to bypass it with a specifically crafted deep link to the Shortcuts app that leverages the `x-cancel` and `x-error` query parameters to open arbitrary URLs when the shortcut isn't successful. Apple fixed it due to my report on 29 July 2025.

CVE-2024-9956 (February 2025)

All major mobile browsers were found to be vulnerable, allowing `FIDO:/` intents to be triggered by a page. As a result a nearby attacker who lures a victim to their site and gets them to accept a passkey prompt can hijack the victim's app session. See the [original blog post](#) for more details.

How CVE-2024-9956 got fixed

All fixes involved blocklisting such URIs to prevent navigation.

Blocklisting is a weak Security Control

Blocklisting is often ineffective as a sole security measure because it only addresses known threats, leaving systems vulnerable to new or unknown attacks. It requires constant maintenance and can create a false sense of security, leading people to overlook other critical defenses. A better approach is to fix the root cause.

Bypass CVE-2024-9956

I found a bypass for the blocklist countermeasure for CVE-2024-9956 in Mobile Safari. The original fix blocks `FIDO:/` URIs from being navigable. However, I discovered a method that allows an attacker to circumvent this restriction and trigger `FIDO:/` intents from attacker-controlled webpages or by simply sending a specifically crafted link to a victim. This bypass enables phishing of FIDO credentials if the physical-presence check can be successfully completed.

The bypass leverages the Shortcuts app, which is installed by default. A specifically crafted deep link to the Shortcuts app uses the `x-cancel` and `x-error` query parameters. These parameters specify a URL that is opened when the interaction is canceled by the user or fails (see [here](#)). Setting these query parameters to a valid `FIDO:/` URL will trigger the intent. This is not blocked and circumvents the current countermeasure for CVE-2024-9956. For example: `shortcuts://x-callback-url/run-shortcut?name=DOESNOTEXIST&x-cancel=FIDO://12345&x-error=FIDO://12345`

Bypass explained step by step

Here is a list of detailed steps:

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The attacker initiates a hybrid (caBLE) FIDO authentication and generates a FIDO intent (`FIDO:/...`) for the target RP on a device within BLE range of the victim's device.

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The attacker lures the victim into opening a link using the previously described `shortcut://` scheme mechanism where the generated FIDO intent is set as the `x-cancel` and `x-error` query parameters. There are various ways to lure a user into opening a phishing link.

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The victim clicks the link, and the FIDO dialog starts.

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The victim authenticates with FIDO.

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The attackers hybrid (caBLE) FIDO authentication completes successfully.

The point is that Safari blocks opening `fido:/` urls but with `shortcut://.....?x-cancel=fido:/` it was still possible to open FIDO intents from webpages rendered in Safari.

Timeline

- 25.03.2025 - Reported to Apple
- 26.03.2025 - First response from Apple
- 29.07.2025 - Fixed in iOS 18.6, iPadOS 18.6, macOS Sequoia 15.6, watchOS 11.6
- 29.07.2025 - Published [Advisory](#) with the comment: "We would like to acknowledge Dennis Kniep for their assistance."

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